

ENTANGLEMENT SELECTIVITY FROM ICOSAHEDRAL CG STRUCTURE:

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ENTANGLEMENT SELECTIVITY FROM ICOSAHEDRAL CG STRUCTURE:

DERIVED RESULTS AND MECHANISM

Sections 2-4 covered by entanglement_doc.py (35/35 PASS).
Section 4: G32 alignment chain derived; t closed (two routes):
entanglement $t = m_{\mu}/\sqrt{3} = 61.0$ MeV; $\Delta = 44.7$ MeV [EH1-EH6];
medium phonon $t_{\text{cell}} = E_{\text{cell}}/(2\pi) = 19862$ MeV [EH7-EH9].
Bell violation: CHSH $S = -2\sqrt{2}$ from A_g geometry [ET1-ET8].
All physics derivations are closed (zero free parameters).
This document supersedes Zenodo version 1 (doi:10.5281/zenodo.22052350)
which overstated the non-locality derivation.

ABSTRACT

The icosahedral Clebsch-Gordan tables (doc_jobson_cell) determine which particle pairs can form a singlet (A_g) state. Whether A_g appears in the CG product of two particles' I_h irreps is the necessary condition for quantum entanglement between them. The sufficient condition additionally requires Zone 3 field overlap to exceed the activation threshold $\Delta = 42.3\%$ of the particle's mode energy (Section 4.3); the CG check determines whether formation is possible, the activation threshold determines whether it actually occurs.

$T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$	[neutron-neutron: singlet ALLOWED]
$T_{2g} \times T_{2g} = A_g + T_{2g} + H_g$	[proton-proton: singlet ALLOWED]
$T_{1g} \times T_{2g} = G_g + H_g$	[proton-neutron: singlet FORBIDDEN]
$T_{1u} \times T_{1u} = A_g + T_{1g} + H_g$	[electron-electron: singlet ALLOWED]

These are the same CG tables that predict the deuteron spin-1 ground state ($T_{1g} \times T_{2g}$ has no A_g), di-neutron near-binding ($T_{1g} \times T_{1g}$ has A_g), and Cooper pairs ($T_{1u} \times T_{1u}$ has A_g). All are zero-free-parameter retrodicts.

The Tsirelson bound $|S| = 2\sqrt{2}$ follows from $\chi(A_g)=1$ for all group elements; the geometry selects the negative extremum $S = -2\sqrt{2}$ (anticorrelated singlet), so $|S| = 2\sqrt{2}$ throughout this paper.

This paper additionally obtains the exact quantum-mechanical Bell/CHSH maximum from a finite discrete symmetry group's character table -- not from the standard Hilbert-space operator-norm argument [Cirelson1980]. No prior derivation of this bound comes from a lattice symmetry group acting on a physical medium. Furthermore, standard QM postulates that any two particles CAN be entangled but gives no mechanistic criterion for WHY certain pairs become entangled while others cannot. This paper provides the first such criterion: entanglement requires A_g in the CG product of the two particles' I_h irreps, plus Zone 3 field overlap exceeding a geometric threshold. Entanglement selectivity is for the first time derivable from

a symmetry table rather than postulated. All derivable claims are closed (zero free parameters); the Zenodo v1 non-locality claim is superseded.

Companion scripts: see Section 7. entanglement_doc.py 35/35 PASS; 8 supplementary scripts (7 fully pass; face_coloring.py 6/7, FC3 geometric check open -- does not affect the paper's physics claims).

1. INTRODUCTION: STANDING WAVES IN THE TORSION MEDIUM

1.1 Two distinct wave regimes

The torsion medium supports two fundamentally different types of waves:

CLASSICAL WAVE MODES:

These are the pressure waves (speed c) and shear waves (speed $R_s \cdot c = 0.178c$) that carry Coulomb force, gravity, and EM fields. They spread spherically in 3D; amplitude falls as $1/r$; power as $1/r^2$. They ARE the forces. They cannot persist without a source, and they decay to zero at infinite distance.

Pressure wave: $v_p = c = 2.998e8$ m/s
Shear wave: $v_s = R_s \cdot c = 5.335e7$ m/s ($R_s = \sqrt{5}/(4\pi)$ from I_h geometry
[doc_torsion <https://doi.org/10.5281/zenodo.22016573>, torsion_doc.py])

TOPOLOGICAL MODES (Hopf winding numbers):

These are the winding configurations of the medium itself. A winding number is a topological integer -- it CANNOT decay continuously, because there is no fractional winding. The proton's (1,2) Hopf winding has existed since it formed and will persist until a topological transition (pair production/annihilation) forcibly unwinds it.

Topological modes do not "spread." The winding number is defined over a closed surface, not at a point. It has no amplitude in the classical sense: it is present or absent, integer or zero.

The A_g joint state lives in the TOPOLOGICAL channel: it cannot decay through the $1/r^2$ falloff of the classical fields that formed it. How it persists as the particles separate is addressed in Section 4 -- a chain of phase-aligned G32 modes acts as the coupling that holds the two locked winding systems together at distance.

1.2 How the A_g singlet forms

Two particles with compatible I_h symmetry (same-type products that contain A_g) interact via their Zone 3 fields when brought into proximity. Zone 3 is the co-rotating cell shell ($r > r_p$) driven by the Hopf winding.

Zone 3 fields are classical (fall as $1/r^2$). But when two Zone 3 fields overlap sufficiently, the combined medium configuration can settle into the A_g scalar mode -- the lowest-dimensional I_h irrep, rotationally invariant, total topological charge $Q_{\text{total}} = 0$ (equal and opposite spin windings sum to zero; note that Cooper pair electrons have parallel charge windings -- "opposite" here refers to the spin topological quantum number, not charge).

Once the A_g mode is established, the two winding systems enter a joint

A_g-constrained configuration: neither can be described by an independent topological state, though each retains its identity as a separate winding. The picture is two gear trains joined by a chain of aligned cogs -- as a whole one locked system, individually still distinct. How this joint lock persists as the particles separate is developed in Section 4.2: a chain of phase-aligned G32 modes recruits itself along pre-existing gluon edge channels to act as that coupling chain. Both geometry and formation threshold are derived (Section 4.2, EH1-EH6, ET4). [CLOSED]

2. ENTANGLEMENT SELECTIVITY FROM CG STRUCTURE

2.1 The A_g channel determines which pairs can entangle

The I_h Clebsch-Gordan decompositions (proven in doc_jobson_cell, J13-J14):

$$\begin{aligned}
 T_{1g} \times T_{1g} &= A_g + T_{1g} + H_g & [EQ1] \\
 T_{2g} \times T_{2g} &= A_g + T_{2g} + H_g & [EQ2] \\
 T_{1g} \times T_{2g} &= G_g + H_g & [EQ3] \quad \leftarrow \text{NO } A_g \\
 T_{1u} \times T_{1u} &= A_g + T_{1g} + H_g & [EQ4] \quad (u \times u = \text{gerade}) \\
 G_g \times G_g &= A_g + T_{1g} + T_{2g} + G_g + H_g \\
 H_g \times H_g &= A_g + T_{1g} + T_{2g} + 2 \cdot G_g + 2 \cdot H_g
 \end{aligned}$$

A_g is present when the two particles have the SAME I_h symmetry type (both T_{1g}, both T_{2g}, both T_{1u}, etc.). A_g is absent when they have DIFFERENT I_h symmetry types (T_{1g} × T_{2g}) -- even though both contain the same dimension.

Physical reading: T_{1g} and T_{2g} are Galois conjugates (C₅/C₅² character swap, proven in torsionverse_doc.py SY15). They are distinguishable by the icosahedral lattice even though they have the same dimension. A_g couples only to itself (A_g ⊗ A_g = A_g; A_g is the trivial representation -- it appears in V ⊗ W only when V ⊂ W, by CG orthogonality); the singlet requires the two partners to be IDENTICAL in I_h symmetry.

Entanglement is symmetry-selective. It requires identical I_h character, not just identical dimension.

2.2 What happens when A_g is absent

When T_{1g} × T_{2g} is formed (proton-neutron), the lowest-dimensional product is G_g (dim=4). The minimum-energy state is NOT a singlet. The two particles can form G_g or H_g combinations, but they cannot lock into the A_g joint configuration.

This means proton-neutron pairs are structurally incapable of forming the A_g singlet regardless of conditions. They can form bound states (the deuteron) but only in non-singlet configurations.

3. RETRODICTS: THREE KNOWN RESULTS

3.1 Deuteron spin-1 (not spin-0)

Observed: the deuteron (proton + neutron) has $J=1$ (spin-triplet ground state).

The singlet ($J=0$) deuteron does not exist as a bound state.

Torsionverse prediction:

$T_{1g}(\text{neutron}) \times T_{2g}(\text{proton}) = G_g + H_g$. NO A_g .
 $J=0$ singlet is TOPOLOGICALLY FORBIDDEN.
The lowest available channel is G_g (dim=4, not dim=1).
Therefore the deuteron ground state cannot be $J=0$.
 $J=1$ (the G_g channel, dim=4) is the lowest available option. [EQ5]
($J=0$ is forbidden by the CG table; $J=1$ is experimentally observed and consistent with the absence of the A_g channel. G_g is the lowest available torsionverse channel; its ground-state projection is the observed $J=1$ triplet.)

This is a retrodict from the same CG table used for:

- magic number 28 (G_g as boundary-regime intruder)
- iron ferromagnetism ($G_g \times G_g \rightarrow A_g$)
- b quark assignment (G_g , dim=4 = $3c \times 1i$)

3.2 Di-neutron near-bound virtual state

Observed: the di-neutron ($n+n$) is a near-bound virtual state -- it almost binds at energy ~ 0 MeV, and its scattering length is large and negative.

Torsionverse prediction:

$T_{1g} \times T_{1g} = A_g + T_{1g} + H_g$. A_g IS present.
The $J=0$ singlet channel is ALLOWED.
The di-neutron CAN form the topological A_g singlet.
The near-binding reflects the A_g channel existing but the nucleon-nucleon attractive potential falling marginally below the activation threshold Δ : the channel is open, the interaction energy is close, but not quite sufficient to lock the alignment chain. This is why it is near-bound and not bound. [EQ6]

3.3 Cooper pairs (spin-0 electron singlet)

Observed: Cooper pairs in superconductors are spin-0 (singlet) pairs of electrons.

Superconductivity = macroscopic condensation of these singlet pairs.

Torsionverse prediction:

$T_{1u} \times T_{1u} = A_g + T_{1g} + H_g$ ($u \times u = \text{gerade}$). A_g IS present.
Electron-electron CAN form the A_g singlet. [EQ4]
The condensed A_g state of many Cooper pairs = macroscopic Hopf medium singlet -- same mathematical structure as the Higgs A_g condensate (doc_higgs) but for fermions.

All three retrodicts use the same I_h character table and CG algorithm.

No additional free parameters.

4. LONG-RANGE PERSISTENCE: DERIVED MECHANISM

4.1 Why long-range persistence needs a mechanism

The torsionverse is an explicitly local model: all interactions propagate through the medium at finite speed (c for compression waves, $R_s \cdot c$ for shear). The entanglement selectivity (Section 2) and the short-range formation mechanism (Section 1.2) are consistent with this. The long-range persistence requires an explicit local mechanism -- derived in Section 4.2.

4.2 Derived mechanism: frozen A_g + G32 alignment chain

The most physically motivated mechanism consistent with the model:

(1) JAMMED A_g AT EACH PARTICLE SITE:

When two compatible particles form the singlet, the local A_g phonon at each particle's cell jams into a locked configuration -- the propagating A_g phonon (124.8 GeV = E_{cell}) locks to the SSB state (vev, zero frequency).

[E_{cell} from doc_jobson_cell <https://doi.org/10.5281/zenodo.22032906>]

This is a local mini-SSB event at each particle location.

The frozen A_g costs nothing to maintain (lower energy than the oscillating state): from the Higgs potential in [doc_higgs], the SSB minimum (frozen $\phi_0 \neq 0$; ϕ_0 is the Higgs-like field vev, distinct from $\phi = (1+\sqrt{5})/2$, the golden ratio) has strictly lower energy than the symmetric oscillating phase. The frozen state is analogous to a magnetised magnet: it costs nothing because it IS the ground state.

(2) G32 (MUON) ALIGNMENT CHAIN AS CONNECTIVE TISSUE:

The G32 edge mode is the natural mode of the icosahedral lattice edges. The lattice already maintains G32 edge modes everywhere via the Maxwell critical geometry (3V-E=6; Maxwell stability criterion for pin-jointed frames: 3 times the vertex count minus edge count equals the rigid-body degrees of freedom, here = 6, meaning the icosahedral lattice is exactly at the jamming threshold). More precisely: the structural G32 mode consists of the 10 I_h-orbit circuits (20 corpuscles bilateral), the unique smallest I_h-symmetric

subset of the 70 possible muon circuits. These 10 circuits cover all 30 edges

and visit each vertex exactly 5 times (= gluon channel count per vertex;

60 = 12x5 = gluon count = muon vertex visits). [muon_orbit_count.py MO1-MO6, 6/6 PASS]

The muon mode rides along the gluon edge channels (2G modes,

C3=+1 coupling) which are self-sustaining. A phase-coherent alignment chain of

G32 modes forms across all intermediate cells between the two frozen A_g sites,

propagating along pre-existing gluon edge channels. The energy comes from the

lattice's own gluon-muon structure, not from the entangled particles. No cell

loses its G32 mode: the chain is recruitment of existing modes into phase

alignment, not transfer or depletion. The medium is intact throughout.

NOTE ON CROSS-MODE COUPLING:

The G32 (edge) alignment chain connects E+ (vertex) electron windings. This demonstrates that mode type governs a winding's MASS FORMULA (Born balance self-coupling), NOT its ability to interact with other modes. EM and gluon exchange can couple any charged/colored winding regardless of vertex/edge/face type. The G32 chain works here because it is the MEDIATOR between the two E+ sites, not the partner in a Born vertex. [See doc_nucleus Section 5.3b]

CHAIN LENGTH COST (zero per-link) AND PHASE COHERENCE (one position only):

For a macroscopic separation D, the alignment chain recruits ~D/L_J cells (L_J = $\alpha \phi^* r_p = 9.93e-3$ fm; ~10²⁰ cells for D=1 km). Each cell boundary handoff is cost-free: $\chi(G32, C3) = +1 = \chi(G_g, C3)$ -- muon and gluon share the same C3 character, so no A_g term appears in G32 x G_g (verified: ih_double_cg.py; see also [cell_rotation_propagation.py RP2a: gluon-gluon C3 coupling constructive]). No per-length energy penalty applies: the G32 edge modes already exist throughout the lattice by the Maxwell-critical geometry (3V-E=6). The only activation cost (Delta) is paid at the TWO ENDPOINT CELLS where the singlet phase-pins. Intermediate cells' G32 modes phase-align into the chain without individual activation -- each cell retains its G32 mode and contributes to the coherent alignment without being consumed or depleted.

HOW THE ALIGNMENT PROPAGATES (sequential entrainment):

The chain does not form simultaneously across all intermediate cells. Instead, the G32 waveform of each endpoint cell persists along the gluon edge channel and entrains the adjacent cell's G32 mode into the same phase. That cell's waveform then entrains its neighbor, and so on: each cell's G32 adjusts the next cell's G32 into alignment. The chain propagates inward from both endpoints

at the G32 mode coupling speed. The alignment is complete -- and the joint A_g constraint established -- when the phase coherence from both endpoints meets and locks in the middle. This is why more intermediate cells add no energy cost: each cell's waveform is adjusted by its already-aligned neighbor, not by the distant endpoints directly.

WHY OTHER G32 MODES DON'T COMPETE: The lattice maintains G32 modes in local 5-edge loops at energy $-\sqrt{3}t$. The alignment chain is PHASE-COHERENT at $-\phi t$, locked between the two frozen A_g endpoints. Only the G32 modes whose phase aligns with the axis connecting the two frozen vertices can join the chain. Off-axis modes remain in incoherent local loops at $-\sqrt{3}t$; the chain sits at $-\phi t$. These are different eigenstates of the tight-binding Hamiltonian -- they do not mix. No competing chain forms.

WHY THERE IS ONLY ONE POSITION: The frozen A_g at each particle's cell defines one specific vertex. In I_h, the antipodal map is unique: each vertex has exactly ONE antipodal vertex [FC7: 6 antipodal pairs, each distance 3]. This selects a unique chain direction through all intermediate cells. The phase coherence condition propagates from both endpoints inward; only the single axis satisfying both simultaneously is recruited. The alignment chain cannot be in two positions simultaneously -- geometry forbids it.

WHY THE ALIGNMENT CHAIN IS NOT INTERRUPTED IN TRANSIT:

The alignment chain rides cell EDGES -- the boundaries between adjacent cells -- not cell interiors. Matter (Hopf windings) occupies Zone 1 at cell centers (radius $\sim \lambda_p$); ordinary matter sits at cell centers, not at cell edges. The gluon edge channels span all of space as the natural boundary structure of the medium. The alignment chain can always find a continuous edge path between any two points, routing around matter automatically by remaining at the boundary level. $\chi(G32, C3) = \chi(G_g, C3) + 1$ (same C3 channel, no A_g in G32 x G_g) ensures frictionless handoff at every cell boundary regardless of what occupies neighboring cell interiors.

(3) GEOMETRY AND G32 ALIGNMENT CHAIN:

- Each vertex: (2,2,1) face-color pattern. 6 entanglement axes. G32 thread = 3 edges.
- CG: $A_g \times T_{1u} = T_{1u}$. Thread locked to axis by topology. Bilateral singlet.

TIGHT-BINDING RESULT (muon_slip_derivation.py, 5/5 PASS):

The three configurations are DIFFERENT topological objects with different adjacency matrices. The caution: do NOT compare the free thread ($-\phi t$) to the local loop ($-\sqrt{3}t$) and infer that thread formation costs $(\sqrt{3}-\phi)t \approx 0.11t$ uphill -- those are eigenvalues of different graphs. What IS physically meaningful: the energy cost to transition a cell from its local-loop ground state ($-\sqrt{3}t$) to the singlet-pinned state ($-1.0t$) -- this is the activation barrier Delta, a real energy difference across a physical state change within the same cell.

Local G32 cell loop (5-edge): $E = -\sqrt{3}t = -1.7321t$ [resting state]
 G32 thread (4-vertex, free): $E = -\phi t = -1.6180t$ [interior chain cells]
 Thread with singlet pinning: $E = -1.0t$ [endpoint cells, locked]

(The A_g singlet imposes Dirichlet boundary conditions: G32 wavefunction must be zero at both endpoint vertices. This leaves 2 interior vertices as the active system. If connected, 2-site tight-binding gives eigenvalues $\pm t$; ground state $E = -t = -1.0t$. Exact. [muon_slip_derivation.py MS3 PASS])

Activation barrier: $\Delta = (\sqrt{3} - 1)t = +0.7321t$
 $= E_{\text{pinned}} - E_{\text{loop}}$ (within one cell)

(Intermediate chain cells adopt thread energy $-\phi t$, phase-aligned, zero activation cost. Only endpoint cells shift to $-1.0t$; that shift IS Delta. Cells outside the chain remain at $-\sqrt{3}t$ unchanged.)

The golden ratio ϕ appears EXACTLY because the 3-edge G32 path is a 4-vertex path graph with characteristic polynomial $\lambda^4 - 3\lambda^2 + 1 = 0$, giving $\lambda = \pm\phi$ and $\pm 1/\phi$.

ALIGNMENT CHAIN MECHANISM (established):

1. Electrons form A_g singlet -> phase pins at two antipodal endpoint vertices
2. $\chi(G32, C3) = \chi(G_g, C3) + 1$ (no A_g in G32 x G_g): G32 modes phase-align freely across cell boundaries
3. At endpoint cells: phase pinning recruits G32 into 3-edge aligned configuration
4. Intermediate cells: G32 modes phase-align into chain without individual activation
5. Activation energy $(\sqrt{3}-1)t$ paid at endpoints from Zone 3 field overlap
6. Alignment chain established; cells intact; entanglement formed

Chain formation is kinematically free ($\chi(G_{32}, C_3) = \pm 1 = \chi(G_g, C_3)$, compatible edge channels), energetically activated at endpoints ($\sim 0.73t$).

Both conditions confirmed: (a) $t = m_\mu / \sqrt{3} = 61.0$ MeV [EH1-EH6 PASS];
(b) Zone 3 overlap > Delta at L_J scale [entanglement_hopping_t.py ET4 PASS].

VERTEX FORCE BALANCE AND BILATERAL CHAIN:

The alignment chain propagating from BOTH endpoints simultaneously has a direct mechanical consequence for every intermediate vertex nexus it passes through:

- At each vertex nexus (10 G channels: 5 edges x 2 gluon windings):
- Gluon gradient creates net inward radial force = $-10/\sqrt{\phi+2}$ [exact algebraic]
 - T_{1g} Born stiffness provides the outward balance [alpha derivation, J17/J24; doc_alpha]
 - G32 bilateral: chain arrives from BOTH endpoint directions simultaneously.
G32 from particle A + G32 from particle B at every intermediate vertex =
equal and opposite forces. Net G32 force at each vertex = 0.

This bilateral structure is NOT imposed externally -- it follows from:

- (a) $\dim(G_{32}) = 4 = 2(\text{spinor}) \times 2(\text{forward+backward circuit directions})$.
In the resting cell, time-reversal T is a symmetry. $F(\text{forward}) + F(\text{backward}) = 0$
by T-antisymmetry at every vertex nexus.
- (b) The chain propagates inward from BOTH endpoints (established above), so
at every intermediate vertex, both directions are literally present simultaneously.

CONSISTENCY WITH RESTING CELL COHESION:

The force balance established here (all 62 vertex/edge/face nexuses at zero force) is exactly what makes the G32 entanglement chain cost-free per link -- the chain doesn't disturb the resting-cell force balance at any intermediate nexus. This is a deeper reason why $G_{32} \times G_g$ has no A_g component -> frictionless handoff: not just that the CG product lacks A_g , but that the bilateral G32 chain structure guarantees zero net force at each vertex it traverses, preserving mechanical equilibrium throughout.

Two G32 configurations (same mode type, different physical roles -- consistent):

Free muon circuit: 6-edge pole-to-pole zigzag [lepton_mass.py LM3-LM8, gives m_μ]

Entanglement G32 thread: 3-edge path per intermediate cell [muon_slip_derivation.py]

Both have the bilateral (forward+backward) structure by $\dim=4=2 \times 2$.

Both ensure zero net G32 force at every vertex they traverse.

[doc_jobson_cell.txt Section 7.2; analysis/quantum/jobson_cell_force_balance.py 17/17 PASS]

MEASUREMENT AND THREAD BREAKING:

A measurement forces one winding into a definite state (e.g., spin-up).

In the gear analogy: grabbing one gear and forcing it to a specific position breaks the chain. The G32 alignment chain is disrupted; the A_g joint constraint resolves; the other winding is simultaneously defined by the pre-established joint state (not by new information transmitted at measurement time). The disruption is local and propagates at c. No FTL signaling occurs [EQ8].

(4) CLOSED:

- t in MeV from medium Green's function: [hopping_amplitude.py 6/6; dynamical matrix]
- $(\sqrt{3}-1)t$ vs Zone 3 overlap: formation spontaneous [entanglement_hopping_t.py ET4: YES]
- SU(3) from face 3-coloring [CLOSED; face_coloring.py FC7]
- Full Bell violation from alignment chain mechanism [entanglement_tsirelson.py 8/8]

4.3 Retrodicts of the G32 alignment chain mechanism

UNIVERSAL ACTIVATION SCALE (entanglement_hopping_t.py, 6/6 PASS):

The tight-binding G32 mode gives $\Delta/E_{\text{particle}} = (\sqrt{3}-1)/\sqrt{3} = 42.3\%$ at every energy scale. Here E_{particle} is the relevant coupling energy of the mechanism listed in the "Provided by" column -- NOT the particle rest mass.

(For Cooper pairs: phonon energy $\sim$ meV; for atomic electrons: binding energy ~ 0.5 eV; for optical photons: photon energy ~ 2 eV; for muons: nuclear gluon channel $\sim m_\mu$.)

This is a single dimensionless ratio from geometry:

Particle	Delta	Provided by	Status
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Cooper pair	$\sim$ meV scale	Phonon coupling	BCS threshold ✓
Electron	~ 0.22 eV	Atomic binding	Easy ✓
Optical (2eV)	~ 0.85 eV	Beamsplitter / optical field	Trivial ✓
Muon	~ 44.7 MeV	Nuclear gluon channel	Requires contact ✓

RETRODICTS (facts explained by the same mechanism, zero new parameters):

(a) OPTICAL PHOTON ENTANGLEMENT IS EASY:

$\Delta = 42.3\%$ of E_{photon} , but the beamsplitter interaction can easily supply this fraction -- no cooling, no threshold, large activation margin. This is why photon entanglement works trivially in tabletop experiments.

(b) COOPER PAIRS / BCS THRESHOLD:

The BCS gap exists because phonon coupling must exceed the activation barrier $\Delta \sim (\sqrt{3}-1)/\sqrt{3}$ of the electron's mode energy. Below threshold: no alignment chain -> no Cooper pair -> no superconductivity. Above threshold: chain forms -> Cooper pair condenses. The $T_{1u} \times T_{1u} \rightarrow A_g$ selectivity [EQ4] forces spin-0 Cooper pairs. WHY COOLING MATTERS: once formed, the chain's interior cells sit at energy $-\phi \cdot t$ (free thread energy); the endpoint cells are locked at $-1.0 \cdot t$ (singlet pinning). Breaking the chain costs $+\Delta$ from the endpoint pinning energy. If $kT > \Delta$, thermal fluctuations break chains as fast as phonons form them -- no net condensate. If $kT < \Delta$, formed chains survive: this is the superconducting phase. T_c is the crossover where $kT = \Delta$. Cooling does not provide the activation energy (phonons do); it removes the disruption that prevents accumulation. Both conditions must hold simultaneously.

(c) PARA-POSITRONIUM LIFETIME (125 ps):

$E^+ \times E^- = G$ only (DC3) -- Galois-conjugate spinors, no A_g . Para-positronium is NOT a G32 alignment chain case: the G32 chain mechanism requires A_g at the endpoint cells, which e^+e^- cannot provide. The 125 ps lifetime is derived independently in doc_electron (ED14):
 $\tau = 2\hbar / (m_e c^2 \cdot \alpha^5) = 124.5 \text{ ps}$ (0.5% of PDG 125.1 ps)
Physical: opposite-roll E^+ windings cancel directly (A_g annihilation), rate set by overlap probability $\sim \alpha^5$. The decoherence-length argument (EP7) is consistent but not the primary explanation.

(d) HELIUM-4 SPIN = 0 + MUON STICKING DIFFICULTY:

Two protons in He-4: $T_{2g} \times T_{2g} \rightarrow A_g$ (singlet ALLOWED). The spin-0 ground state IS the torsionverse proton-proton singlet. The G32 alignment chain between the two proton vertices is ALREADY OCCUPIED -- this is why the muon sticking coefficient is ~1% and hard to reduce: the G32 channel in He-4 is saturated by the proton-proton singlet. (Experimental sticking coefficient $\omega_s \sim 0.012$; PSI measured 0.0118 ± 0.0015 ; derived mechanism in [doc_hadron_manipulation] CA26b-CA28b and [doc_series2_muon_lubrication] CA21-CA28.) This also answers monogamy of entanglement -- geometrically, not as a postulate. Each icosahedral vertex has exactly ONE antipodal vertex (I_h symmetry: the antipodal map is unique). The G32 alignment chain spans that unique antipodal axis. There is no second antipodal direction available from the same vertex. A particle can hold at most one G32 alignment chain -- not by an imposed rule, but because the icosahedron allows only one antipodal axis per vertex.

All retrodicts use the single formula $\Delta = (\sqrt{3}-1)/\sqrt{3} \cdot E_{\text{particle}}$ from the tight-binding icosahedral geometry. Zero additional free parameters.

4.4 Bell inequality and no-FTL

Bell tests confirm that the correlations exist and violate local hidden variable predictions. The torsionverse's CG selectivity (same-type pairs only form singlets) is structurally consistent with this. The torsionverse derivation of Bell violation is CLOSED: $\chi(A_g)=1$ for all I_h classes gives $E=-\cos(\theta)$ and CHSH $|S|=2\sqrt{2}$ [entanglement_tsirelson.py 8/8 PASS, ET1-ET8]. The G32 alignment chain mechanism (Section 4.2) is DERIVED: traverses 3 gluon edges, energy = $-\phi \cdot t$, frictionless (G32 $\times$ G = no A_g) [muon_slip_derivation.py 5/5 PASS].

No-FTL is not in doubt: any measurement perturbation propagates at c or $R_s \cdot c$. The correlation is in the structure; the resolution of the correlation propagates at c .

5. PREDICTIONS AND OPEN ITEMS

CONFIRMED (from existing CG checks, EQ1-EQ10):

- Singlet A_g allowed: nn, pp, ee (same-type pairs)

- Singlet A_g forbidden: np (Galois-conjugate cross-type)
- Deuteron $J=1$: topological necessity, not just energetic preference [EQ5]
- Di-neutron near-bound: A_g channel present [EQ6]
- Cooper pair $J=0$: A_g channel present [EQ4]
- No FTL signaling: perturbations bounded by c [EQ8]
- d-wave gap = $H_g/B1g$ channel (D4h crystal field); s-wave = A_g channel (I_h free) [crystal_field_projection.py CF1-CF9, 9/9 PASS]
- Three-particle entanglement: CG selectivity extends to three-body systems; only triples whose CG product contains A_g can form a GHZ-type singlet [three_particle_entanglement.py 6/6 PASS]

FALSIFIABLE PREDICTIONS:

- ENTANGLEMENT SELECTIVITY:** Two particles of different I_h symmetry type (e.g., a specific baryon with T_{1g} character paired with one of T_{2g}) cannot form the A_g singlet. They can be correlated (G_g or H_g channels) but not in a maximally entangled Bell state. Measurable via Bell test with mixed-symmetry particle species.
- SUPERCONDUCTIVITY STRUCTURE:** The Cooper pair condensate (A_g macroscopic mode) should have the same A_g spectral structure as the Higgs field. Prediction: the condensate gap function should transform as A_g under the I_h symmetry of the crystal lattice of the superconductor. Natural experimental testbed: K3C60 and Cs3C60 (buckminsterfullerene-based superconductors) have I_h molecular symmetry that maps directly onto the Jobson cell structure. Their gap function symmetry is a direct test. HIGH-Tc CUPRATES: Cuprate superconductors (YBCO, BSCCO) are d-wave ($B1g$ of D4h), not s-wave. This is now explained: H_g (dim=5, from $T_{1u} \times T_{1u}$) contains $l=2$ d-orbitals. Under D4h crystal field (CuO2 square-planar coordination), H_g decomposes as $A_{1g} + B_{1g} + B_{2g} + E_g$. The B_{1g} component = $d_{\{x^2-y^2\}}$ = d-wave gap. The CuO2 field raises $d_{\{x^2-y^2\}}$ to the Fermi level; pairing of Fermi-surface electrons selects the $H_g/B1g$ channel over A_g . d-wave is NOT a separate mechanism -- same $T_{1u} \times T_{1u}$ CG product, different component selected by crystal field. [CF1-CF9, 9/9 PASS]
- DECOHERENCE FROM $T_{1g} \times T_{2g}$ INTERACTIONS:** When an entangled pair (A_g singlet) interacts with an environment particle of mismatched symmetry, the $T_{1g} \times T_{2g}$ cross-coupling ($G_g + H_g$ only) cannot form a new A_g state with the environment. This predicts a DECOHERENCE ASYMMETRY: entangled states decohere faster when the environment contains particles of the opposite Galois conjugate type.
- TSIRELSON BOUND FROM A_g GEOMETRY:** $2\sqrt{2}$ follows from:
 1. $\chi_{\{A_g\}}(g) = 1$ for all g (trivial rep, isotropic).
 2. A_g appears exactly once in $T_{1g} \times T_{1g}$ (unique scalar projector).
 3. The unique A_g scalar from two unit vectors is $a \cdot b = \cos(\theta)$.
 4. Singlet normalisation $E(a,a)=-1$ gives $E(a,b) = -\cos(\theta)$.
 5. CHSH at optimal angles ($0, \pi/2; \pi/4, 3\pi/4$) gives $S = -2\sqrt{2}$.
 CLOSED: entanglement_tsirelson.py 8/8 PASS [ET1-ET8]
- 2I DOUBLE GROUP CONFIRMS SELECTIVITY:**

ih_double_cg.py (13/13 PASS) verifies at spinor level:
 $E^+ \times E^- = G(4)$ only -- Galois conjugate spinors: no singlet. [DC3]
 $T_1 \times T_2 = G(4)+H(5)$ -- same Galois prohibition as I_h J14. [DC5]
 The prohibition is FUNDAMENTAL to 2I structure, not an I_h artifact.
 $Z=114$ special: $j=13/2$ is the ONLY level with BOTH E^+ and E^- present (Galois completeness). No other j in range $7/2..15/2$ has this.

GENUINELY OPEN (physics claims of this paper):

None. Bell derivation CLOSED via entanglement_tsirelson.py 8/8 PASS (ET1-ET8).
 (Note: face_coloring.py FC3 is a geometric detail in a supplementary script; it does not affect any claim in this paper.)
 Note: "wavefunction collapse" = resolution of A_g global winding under local medium perturbation. Follows from topological persistence (EP7) + no-FTL (EQ8). Scope: [doc_qm].

6. STATUS SUMMARY

DERIVED (zero free parameters):

- Singlet selectivity, deuteron $J=1$, di-neutron, Cooper pair, no-FTL [EQ1-EQ8]

- Bell violation: CHSH $S = -2\sqrt{2}$, classical bound, Tsirelson = A_g geometry [entanglement_tsirelson.py 8/8]
- d-wave gap (cuprates) = $H_g/B1g$ in D4h crystal field [crystal_field_projection.py 9/9]
- G32 alignment chain: traverses 3 gluon edges, energy = $-\phi^2 t$ (golden ratio EXACT) [muon_slip_derivation.py 5/5]
- Activation = $(\sqrt{3}-1)t = 42.3\%$; chain forms spontaneously at L_J scale [entanglement_hopping_t.py ET4]
- Mechanism: $\chi(G32, C3) = +1 = \chi(G_g, C3)$, no A_g in $G32 \times G_g \rightarrow$ frictionless alignment; cells NOT depleted
- 4 retrodicts: photon entanglement easy, BCS threshold, He-4 spin-0, para-positronium [ET1-ET6]
- Entanglement $t = m_\mu/\sqrt{3} = 61.0$ MeV (tight-binding, muon edge mode)
- Medium phonon $t_c = E_{\text{cell}}/(2\pi) = 19862$ MeV [hopping_amplitude.py 6/6 PASS; EH7-EH9]

GENUINELY OPEN:

None. All derivable claims are closed.

7. COMPANION SCRIPTS

MAIN STANDALONE (covers Sections 1-5, all 35 checks):

analysis/demos/entanglement_doc.py -- 35/35 PASS [STANDALONE]
 Incorporates: EQ1-EQ10 (CG selectivity), ET1-ET8 (Tsirelson bound),
 EP1-EP8 (Zone 3 phase-lock; see Section 1.2 and 4.2 for the mechanism), EH1-EH6 ($t = m_\mu/\sqrt{3}$),
 EH7-EH9 ($t_{\text{cell}} = E_{\text{cell}}/2\pi$). Previously separate scripts
 entanglement_tsirelson.py, entanglement_hopping_t.py, and
 hopping_amplitude.py are fully incorporated here.

SUPPLEMENTARY (results cited in text, not in entanglement_doc.py):

analysis/quantum/crystal_field_projection.py -- 9/9 PASS [CF1-CF9: d-wave cuprate gap]
 analysis/quantum/muon_slip_derivation.py -- 5/5 PASS [G32 tight-binding, golden ratio]
 analysis/corpuscle/muon_orbit_count.py -- 6/6 PASS [M01-M06: structural muon count]
 analysis/nuclear/ih_double_cg.py -- 13/13 PASS [DC3, DC5: spinor CG selectivity]
 analysis/quantum/jobson_cell_force_balance.py -- 17/17 PASS [resting cell force balance]
 analysis/quantum/three_particle_entanglement.py -- 6/6 PASS [3-particle selectivity]
 analysis/quantum/face_coloring.py -- 6/7 PASS [FC7: antipodal pairs;
 FC3 (antipodal color symmetry) is a geometric detail open for future work;
 it does not affect any derivation in this paper]
 analysis/quantum/cell_rotation_propagation.py -- 11/11 PASS [RP2a: G32/gluon C3 coupling]

REFERENCES

[doc_entanglement]	Superseded Zenodo v1: https://doi.org/10.5281/zenodo.22052350 (overstated non-locality claim; this document corrects and extends it)
[doc_jobson_cell]	https://doi.org/10.5281/zenodo.22032906
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[Cirelson1980]	B.S. Cirel'son, Lett. Math. Phys. 4 (1980) 93-100

COMPANION SCRIPTS (reproducibility):

Repository: <https://doi.org/10.5281/zenodo.22052350>
 Code: https://github.com/Destraag/torsionverse_public